

Topic: Divergence & Curl

I. Divergence

Defn: Let $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ be a vector-valued function.
The **divergence** of $\mathbf{F}$, denoted $\text{div}(\mathbf{F})$, is given by

$$\text{div}(\mathbf{F}) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = M_x + N_y + P_z$$

Note: This can be remembered as: $\text{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$

where $\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]$ is the **del operator**.

Ex: Let $\mathbf{F}(x, y, z) = [0, \cos(xyz), \sin(xyz)]$. Find $\text{div}(\mathbf{F})$ at $P = (2, \frac{\pi}{2}, 0)$

Since $M_x = 0$, $N_y = -\sin(xyz)(xz)$, $P_z = \cos(xyz)(xy)$

$$\Rightarrow \text{div}(\mathbf{F}) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = -\sin(xyz)(xz) + \cos(xyz)(xy)$$

$$\Rightarrow \text{div}(\mathbf{F})(P) = -\sin(0)(0) + \cos(0)(\pi) = \pi$$

II. Curl

Defn: Let $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ be a vector-valued function.
The **curl** of $\mathbf{F}$, denoted $\text{curl}(\mathbf{F})$, is given by

$$\text{curl}(\mathbf{F}) = \left[\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}, \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}, \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] = [P_y - N_z, M_z - P_x, N_x - M_y]$$

Note: This can be remembered as: $\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F}$

Ex: Let $\mathbf{F}(x, y, z) = [e^{-z^2}, e^{-x^2}, e^{-y^2}]$. Find $\text{curl}(\mathbf{F})$

$$P_y - N_z = (e^{-y^2})_y - (e^{-x^2})_z = -2ye^{-y^2} - 0$$

$$M_z - P_x = (e^{-z^2})_z - (e^{-y^2})_x = -2ze^{-z^2} - 0$$

$$N_x - M_y = (e^{-x^2})_x - (e^{-z^2})_y = -2xe^{-x^2} - 0$$

$$\Rightarrow \text{curl}(\mathbf{F}) = [-2ye^{-y^2}, -2ze^{-z^2}, -2xe^{-x^2}]$$

III. Additional Concepts

Defn: Let $f(x, y, z)$ be a scalar-valued function. The **Laplacian** of f is

$$\operatorname{div}(\nabla f) = \nabla \cdot \nabla f = \nabla^2 f$$

Defn: Let $\mathbf{F}(x, y, z)$ be a vector-valued function. We say $\mathbf{F}$ is an **incompressible** (or **solenoidal**) vector field if

$$\operatorname{div}(\mathbf{F}) = 0$$

Defn: Let $\mathbf{F}(x, y, z)$ be a vector-valued function. We say $\mathbf{F}$ is an **irrotational** vector field if

$$\operatorname{curl}(\mathbf{F}) = \mathbf{0}$$

Note: Irrotational field are **Conservative**.

Thm: Let f be scalar-valued and $\mathbf{F}$ be vector-valued.

- Gradient fields are irrotational: $\operatorname{curl}(\nabla f) = \mathbf{0}$
- The curl of a vector field is incompressible: $\operatorname{div}(\operatorname{curl} \mathbf{F}) = 0$

Ex: Let $\mathbf{v} = [0, 3z^2, 0]$ be a velocity field of some steady fluid flow. Answer the following:

- Is the flow incompressible? (i.e. is $\text{div}(\mathbf{v}) = 0$?)
- Is the flow irrotational? (i.e. is $\text{curl}(\mathbf{v}) = \mathbf{0}$?)

$$\text{div}(\mathbf{v}) = (0)_x + (3z^2)_y + (0)_z = 0 + 0 + 0 = 0 \quad \mathbf{v} \text{ is incompressible.}$$

$$P_y - N_z = (0)_y - (3z^2)_z = -6z$$

$$M_z - P_x = (0)_z - (0)_x = 0$$

$$N_x - M_y = (3z^2)_x - (0)_y = 0$$

$$\text{So } \text{curl}(\mathbf{v}) = [-6z, 0, 0] \neq [0, 0, 0] \implies \mathbf{v} \text{ is not irrotational}$$

Ex: Find the general form for $N(x, y, z)$ for which $\mathbf{F} = [-x \ln z, N, z^2]$ is incompressible?

Want to find N such that $\text{div}(\mathbf{F}) = 0$

That is,

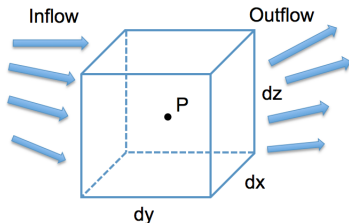
$$0 = \text{div}(\mathbf{F}) = (-x \ln z)_x + N_y + (z^2)_z = -\ln z + N_y + 2z$$

$$\implies N_y = \ln z - 2z$$

$$\implies N = y \ln z - 2yz + g(x, z) \quad (g \text{ can be any function of } x \text{ and } z)$$

IV. Physical Meaning

The Divergence at a point, P , can be thought of as a measurement of much flow goes in the box compared to how much flow goes out of the box (Net flow).



Divergence at $P \sim$ Net flow through a "small" box around P

If the outflow = inflow (net flow out = 0) then

$$\text{div}(\text{flow}) = 0$$

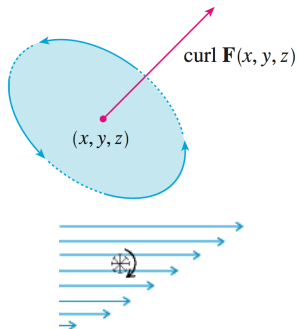
If outflow $>$ inflow (net flow out $>$ 0) then

$$\text{div}(\text{flow}) > 0$$

If outflow $<$ inflow (net flow out $<$ 0) then

$$\text{div}(\text{flow}) < 0$$

The Curl at a point, P , measures the tendency for a flow to "curl" around a point.

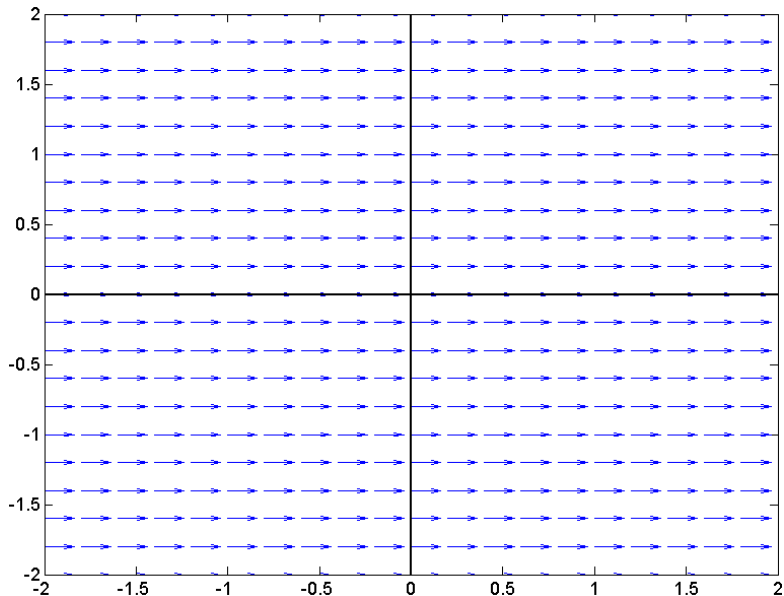


Flow acting counterclockwise around a point produces a curl pointing **out of the board**.

Flow acting clockwise around a point produces a curl pointing **into the board**.

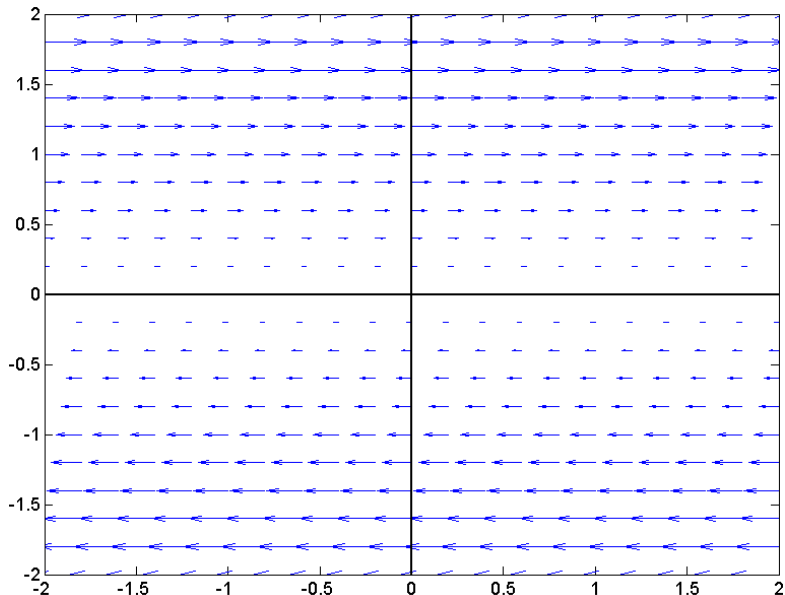
Ex: Let $\mathbf{F}$ be the flow field below. Is the flow...

- ...incompressible? **Yes** If not, then is $\text{div}(\mathbf{F})$ positive or negative?
- ...irrotational? **Yes** If not, then is $\text{curl}(\mathbf{F})$ pointing into or out of the board?



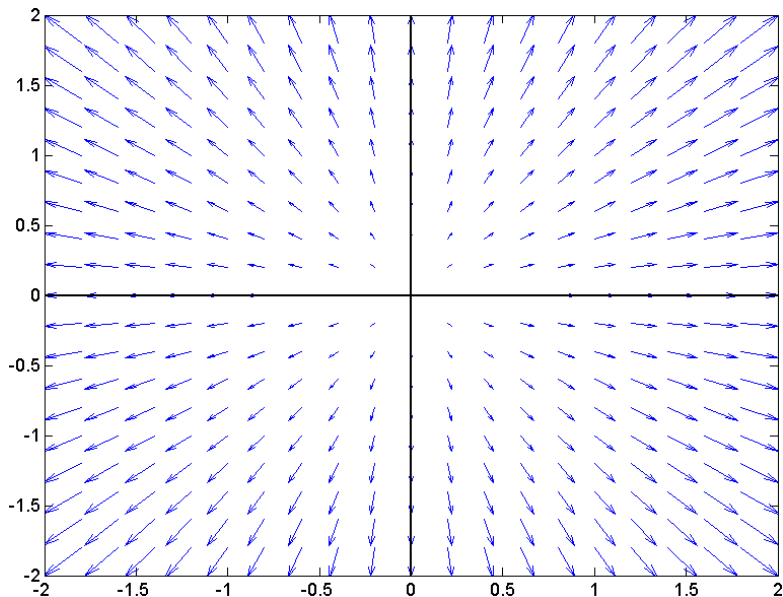
Ex: Let $\mathbf{F}$ be the flow field below. Is the flow...

- ...incompressible? **Yes** If not, then is $\text{div}(\mathbf{F})$ positive or negative?
- ...irrotational? **No** If not, then is $\text{curl}(\mathbf{F})$ pointing into or out of the board?



Ex: Let $\mathbf{F}$ be the flow field below. Is the flow...

- ...incompressible? **No** If not, then is $\text{div}(\mathbf{F})$ positive or negative? **positive**
- ...irrotational? **Yes** If not, then is $\text{curl}(\mathbf{F})$ pointing into or out of the board?



V. Summary

- $\mathbf{F}$ conservative $\iff \mathbf{F} = \nabla f$ for some $f \iff \int_C \mathbf{F} \cdot d\mathbf{r}$ path independent $\iff \text{curl}(\mathbf{F}) = \mathbf{0}$
- $\mathbf{F}$ conservative & C closed $\implies \int_C \mathbf{F} \cdot d\mathbf{r} = 0$
- $\text{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$ (divergence)
- $\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F}$ (curl)
- $\text{div}(\nabla f) = \nabla \cdot \nabla f = \nabla^2 f$ (Laplacian)
- $\text{div}(\mathbf{F}) = 0 \implies \mathbf{F}$ incompressible/solenoidal
- $\text{curl}(\mathbf{F}) = \mathbf{0} \implies \mathbf{F}$ irrotational
- $\text{curl}(\nabla f) = \mathbf{0} \longleftarrow \text{zero vector}$
- $\text{div}(\text{curl } \mathbf{F}) = 0$